

IB MYP
Formative Assessment - 1 (2026-27)

Students Name:	Grade & Sec: MYP-1	Date: -07-2026
Subject: Integrated Sciences	Task Duration: 1 hour	Level Obtained
Unit Title : What do Scientists do?		
Criteria & Strands assessed: Criterion B Students should be able to: i. outline an appropriate problem or research question to be tested by a scientific investigation ii. Formulate a testable prediction using scientific reasoning iii. Explain how to manipulate the variables , and outline how data will be collected iv. Design scientific investigations		
ATL Category, Cluster, Strand:		

Student's Reflection after the assessment:

My strengths in this assessment:	
Which part of the task did I understand clearly and complete confidently?	
How did I effectively use ATL skills in this assessment?	
My weak areas in this task	
Which part of the task did I find most challenging, and what made it difficult for me?	
Skills that I have developed through this task	
What new skills or approaches did I develop or improve while completing this assessment?	
My strategies for further improvement	
What specific actions will I take to improve my performance next time?	

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Teacher's Feedback:

Concepts where you did good :

Needs further practice :

Strand-i	Q-1	Q-2
Strand-ii	Q-3	Q-4
Strand-iii	Q-5	Q-6
Strand-iv	Q-7	
Level Obtained		

Achievement Level	Level Descriptor (Task-Specific Clarification)
0	The student does not reach a standard described by any of the descriptors below.
1-2	The student is able to: i. State a simple research question related to the investigation. ii. State a simple prediction related to the investigation. iii. state one or more variables with limited accuracy. iv. design a simple investigation with limited success by listing a few materials or steps.
3-4	The student is able to: i. state an appropriate research question for the given investigation. ii. state a testable hypothesis related to the investigation. iii. state how the variables will be manipulated and state how data will be collected and recorded. iv. design a safe investigation in which appropriate materials are selected and basic logical

5-6	The student is able to: i. state a clear and focused research question that can be scientifically investigated. ii. outline a testable hypothesis using If...Then...Because... with appropriate scientific reasoning. iii. outline how the variables will be manipulated and state how relevant data will be collected and recorded. iv. design a complete and safe investigation by selecting appropriate materials, identifying variables and describing a logical method.
7-8	The student is able to: i. outline a precise and scientifically testable research question using appropriate scientific vocabulary. ii. outline a testable hypothesis using If...Then...Because... with accurate scientific reasoning. iii. outline how the variables will be manipulated and outline how sufficient, relevant and reliable data will be collected and recorded. iv. design a logical, complete and safe investigation by selecting appropriate materials, identifying variables, describing a fair method and including suitable safety precautions.

Question 1: Shadow Investigation

Students investigate how the position of a torch affects the length of a shadow.

Task:

State what is changing?	
State what is measured?	

Outline a research question that could be tested through this scientific investigation.

2 Sound Investigation

Students investigate how the distance from a speaker affects the loudness heard.

State

Independent Variable	
Dependent Variable	

Outline a research question that could be tested through this scientific investigation.

SECTION B: HYPOTHESIS

Strand ii

Formulate a testable hypothesis **and explain** it using scientific reasoning

Question 3: Seed Germination

A student wants to investigate how the amount of light affects seed germination.

Tasks:

1. **Formulate a testable hypothesis using the format:**

If..., then..., because... Explain your prediction using scientific reasoning.

Question 4: Toy Car Ramp

A student wants to investigate how the height of a ramp affects the distance travelled by a toy car.

1. Formulate a testable hypothesis using the format:

If..., then..., because... Explain your prediction using scientific reasoning

SECTION C: VARIABLES

Strand iii

Explain how the variables will be manipulated and how data will be collected and recorded.

Question 5: Cooling Water

A student investigates how the type of container affects the rate at which water cools.

Containers

- Plastic
- Glass
- Metal

Complete the table.

Independent Variable	
Dependent Variable	
Controlled Variable(s)	

Explain how you will manipulate the independent variable.	
Explain how data will be collected and recorded.	

Question 6: Paper Airplane

A student investigates how the wing length of a paper airplane affects the distance it travels.

Complete the table.

Independent Variable	
Dependent Variable	
Controlled Variable(s)	
Explain how you will manipulate the independent variable.	
Explain how data will be collected and recorded.	

SECTION D: Strand iv-DESIGNING AN INVESTIGATION

Choose **ONE** investigation.

Option A: Seed Germination Investigation

Investigate how the amount of light affects seed germination.

Investigate how the height from which a ball is dropped affects the number of bounces.

- Design scientific investigations and describe a logical method that makes the investigation a fair test.
- Identify the independent, dependent and controlled variables.
- List the materials required.
- Explain how data will be collected and recorded
- State safety precautions.

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